IEEE Access

Supplementary Material for "Reuse, but Verify: Certified Maintenance of Table Cells Computed by Large Language Models under Prompt Edits"

ARJUN LOHAN

University of Southern California, Los Angeles, CA, USA (e-mail: lohan@usc.edu)

This supplement holds the appendices that the article cites by letter. Its tables and figures are numbered S1, S2, and so on. A section, table, figure, equation, assumption, or theorem number without the prefix S is the article's, and citations refer to the article's reference list.

APPENDIX A THE MECHANICAL GUARDS

The checker (`scripts/check-paper-numbers.ts`) fails on any of ten conditions.

1) A hand-typed quantity in the prose outside an enumerated allow-list (user parameters such as α and δ, protocol constants such as the schedule looks, and generic thresholds quoted as such).
2) A quantity hardcoded inside the generator instead of derived from an artifact.
3) A macro defined but never cited, since an uncited macro can drift undetected.
4) A macro whose trailing space is swallowed into the next word, which is invisible in the source.
5) A sentence that asserts two quantities differ while citing two macros whose values are equal.
6) A spelled-out multiplier ("twice", "an order of magnitude") that the cited macros' ratio does not support.
7) A containment claim ("inside", "at most") that the cited range and scalar macros do not support.
8) An absolute claim ("never", "every", "anywhere") appearing in the abstract or the contributions list without an entry in an enumerated allow-list. This is the one class of defect the numeric guards are blind to by construction.
9) A scope claim ("on the same column") whose cited macros come from different columns, checked against a provenance file the generator emits from the same pair definitions the experiments run on.
10) A sentence of the prose, of a list item, of a caption, or of the abstract longer than 60 words (table bodies, displayed mathematics, and the theorem environments excluded), a readability bound.

The generator adds guards of its own. It asserts the relations that Section V, Section VIII, and the appendices state between figures: in the paragraphs on the main results, the strict mode, the stratifier ablation, the matched competitor, the model study, the independence check, the usable budgets, the graded relation, and the pilot. It also asserts that the separate audit verifier reproduces each audit the article prints, and that the released certify action is configured as the pinned procedure. It stops if a regenerated artifact breaks one. It rounds every figure half up on its decimal value, and rounds upward a figure quoted as a maximum ("at most"). Generating the quantities is necessary but not sufficient, because the sentences that *relate* quantities go stale independently of the quantities themselves.

APPENDIX B THE BOUND ABLATION IN DETAIL

a: The bound on the deployment stratum.

In the 81,469-cell deployment stratum, a clean sample of 90 at the pinned per-look level gives an exact bound of 0.052, which coincides with the binomial limit to three digits.

b: The bounds compared.

Over the 20 configurations of Table 2 (main seed), the exact bound's mean savings are 29.5% against 18.5% under Maurer–Pontil and 16.4% under Bardenet–Maillard, on 7,458 oracle calls against 11,515 and 12,325. Relative to the exact bound, Maurer–Pontil loses 3 certificates and gains 0, and of the shared ones it reaches 1 at the same look and 6 at a later one. The widest single gain is SO scope widening at α=0.1, from 18.1% savings for 1,485 calls to 72.1% for 405. Of the two empirical-Bernstein bounds, the without-replacement form pays more, not less, because its constants are larger. The preprint reported it as tighter. Its replay used the Maurer–Pontil linear constant inside the Serfling form, and the published constant, $\kappa = 7/3 + 3/\sqrt{2}$ [14], is what the numbers here use. The Clopper–Pearson limit certifies 9 configurations at mean savings 28.1% on 8,054 calls, the finite-population correction buying the exact bound its one further certificate. The betting sequence certifies 9 configurations at mean savings 29.1% on 7,616 calls. It beats the exact bound on 1 of the 20 configurations, where the schedule's Bonferroni split bites hardest (Djinni criteria change at α=0.2: 76.3% against 62.8%). It loses to that bound on 2 and matches it on the rest (certificates lost 1, gained 0). Its validity here rests on a capital process built for draws with a constant conditional mean, which without-replacement draws lack. The without-replacement variant given in [23], which we did not implement, or a time-uniform boundary [31] would remove the split without that caveat and is the remaining power lever we see.

Validity comes before power, so all five arms were replayed through the calibration protocol, in a run separate from and

TABLE S1. Five upper confidence bounds inside the same pinned procedure on the same stored labels. The first two columns give each bound on a clean sample of 90 draws and on the deployment evidence of 5 flips in 180 draws. The four fixed-sample bounds are read at the pinned per-look level, $\delta/12$, and the betting sequence at the unsplit per-stratum level, $\delta/2$ (Appendix B states the caveat on its validity). The next three columns give the certifying configurations, mean savings, and oracle calls over the Table 2 grid (main seed). The last gives the worst rate of unsafe certificates, the event of Theorem 1, at planted nulls in the calibration replay (per-stratum δ=0.05). WoR: without replacement.

	upper bound on		Table 2 grid (20 configurations)			unsafe at nulls
Bound	$0/90, N{=}1{,}800$	$5/180, N{=}81{,}469$	certifying	mean savings	oracle calls	worst
Exact finite-population (pinned)	0.050	0.073	10	29.5%	7,458	1.00%
Clopper–Pearson (binomial)	0.052	0.073	9	28.1%	8,054	0.50%
Betting confidence sequence	0.065	–	9	29.1%	7,616	1.20%
Maurer–Pontil	0.144	0.140	7	18.5%	11,515	0.00%
Bardenet–Maillard (WoR)	0.317	0.230	7	16.4%	12,325	0.00%

The betting sequence has no $5/180$ entry because its bound is a function of the whole draw sequence, not of (k, n) alone.

smaller than the calibration study of Section V-F (36,000 trials per arm, 18,000 of them at planted rates above the budget). At planted nulls the exact bound certifies up to 6.3% of trials, against a per-stratum δ_j of 5%. Most of those certificates are correct, because the realized count of a planted stratum falls below αn_j in many trials. The certificates whose realized count exceeds the budget, the event of Theorem 1, are at most 1.00% of trials in any null configuration ($\alpha{=}0.1$, $p{=}0.105$, $n_j{=}1{,}800$; Wilson [0.54, 1.83]%). The certification rate at a planted null is not the theorem's event, and it may exceed δ_j without a violation. In 0 of the 18 null configurations does its interval lie wholly above δ_j. The Clopper–Pearson limit certifies 1.5% of null trials at worst (0.50% unsafe), and the betting sequence 2.5% (1.20% unsafe). Maurer–Pontil and Bardenet–Maillard certify 0.0% and 0.0% (0.00% and 0.00% unsafe), leaving almost all of their δ_j unspent. A clean 1,800-cell stratum at $\alpha{=}0.05$ costs the exact bound 90 draws (180 at $n_j{=}5{,}000$ in the calibration study). That figure is a tie: at exactly that size the bound on a clean look of that many draws equals the budget, and a stratum a few cells larger pays the next look. It costs the Clopper–Pearson limit 180, the sequence 180, Maurer–Pontil 360, and Bardenet–Maillard 720. The frontier in Fig. 2 is therefore a property of the bound.

c: Configuration by configuration.

Table S2 gives, for every configuration of Table 2 and each of the five bounds, the certified savings of the main seed and the oracle calls spent. Relative to the exact bound, the Clopper–Pearson bound certifies all but one of its configurations (lost 1, gained 0), reaching 7 of them at the same look, 0 earlier, and 2 later, since it ignores the finite-population correction. Maurer–Pontil reaches 1 at the same look, 0 earlier, and 6 later. Bardenet–Maillard loses 3 certificates, gains 0, and reaches 0 at the same look, 0 earlier, and 7 later. The betting sequence reaches 7 at the same look, 1 earlier, and 1 later.

TABLE S2. Certified savings of the main seed (oracle calls in parentheses) for every configuration and bound; "refused" gives the calls spent before the stop.

Edit	α	exact	Clopper–Pearson	betting	Maurer–Pontil	Bardenet–Maillard
SO formatting-only	0.02	refused (225)	refused (225)	refused (225)	refused (225)	refused (225)
SO formatting-only	0.05	18.7% (1,485)	18.7% (1,485)	18.7% (1,485)	refused (1,485)	refused (1,485)
SO formatting-only	0.10	88.5% (90)	86.2% (135)	86.2% (135)	54.7% (765)	54.7% (765)
SO formatting-only	0.20	88.5% (90)	88.5% (90)	88.5% (90)	86.2% (135)	72.7% (405)
SO synonym rewording	0.02	refused (90)	refused (90)	refused (90)	refused (90)	refused (90)
SO synonym rewording	0.05	refused (90)	refused (90)	refused (90)	refused (90)	refused (90)
SO synonym rewording	0.10	refused (90)	refused (90)	refused (90)	refused (90)	refused (90)
SO synonym rewording	0.20	82.3% (225)	82.3% (225)	82.3% (225)	82.3% (225)	73.3% (405)
SO scope widening	0.02	refused (90)	refused (90)	refused (90)	refused (90)	refused (90)
SO scope widening	0.05	refused (765)	refused (765)	refused (765)	refused (765)	refused (765)
SO scope widening	0.10	72.1% (405)	72.1% (405)	72.1% (405)	18.1% (1,485)	18.1% (1,485)
SO scope widening	0.20	87.8% (90)	87.8% (90)	87.8% (90)	81.1% (225)	72.1% (405)
Djinni formatting-only	0.02	refused (180)	refused (180)	refused (180)	refused (180)	refused (180)
Djinni formatting-only	0.05	refused (315)	refused (315)	refused (315)	refused (315)	refused (315)
Djinni formatting-only	0.10	19.2% (495)	refused (878)	refused (878)	refused (878)	refused (878)
Djinni formatting-only	0.20	52.4% (953)	52.4% (953)	52.4% (953)	19.2% (1,617)	19.2% (1,617)
Djinni criteria change	0.02	refused (180)	refused (180)	refused (180)	refused (180)	refused (180)
Djinni criteria change	0.05	refused (315)	refused (315)	refused (315)	refused (315)	refused (315)
Djinni criteria change	0.10	18.9% (540)	18.9% (540)	18.9% (540)	refused (918)	refused (918)
Djinni criteria change	0.20	62.8% (745)	54.4% (913)	76.3% (475)	27.9% (1,442)	18.9% (1,622)

APPENDIX C THE DEPLOYMENT RUNS IN DETAIL

This appendix gives the evidence behind Section V-E: the materialization, the certificate the system applied in August, the replay of its draws under the exact bound, the drift measurement, the September certification, and the call counts with three reproducibility caveats.

a: The materialization.

The column's 78,956 billed cells cost $9.31 at $0.000118 per cell. The run took 7.2 hours of wall clock at its 32-way concurrency, measured from its first cell write to its last. An estimate from stored latencies agrees: the column's 80,902 computed cells, this run's and those of earlier passes, average 10.5 seconds per call, and those latencies, summed and divided by the concurrency, give 7.4 hours. The remaining 10,228 rows cost nothing: rows with identical bound content share one stored cell, which is why the billed cells are fewer than the rows. The cells this article counts as sampled, reused, or recomputed are rows, and its savings are shares of rows.

b: The applied August certificate.

Theorem 1 does not cover the live August run, which used the Maurer–Pontil bound; it is reported because it is the certificate the system applied (Table S3). Under Maurer–Pontil, at α=0.2 the procedure spent 225 oracle calls to reuse 81,289 cells (91.1% fewer calls). It certified the cached-FALSE stratum at look 180 with 5 flips. Audited error was 3.16% over 2,686 audited reused rows. It refused the 7,715-row cached-TRUE stratum and so left for recomputation the 7,670 of its cells that the futility stop had not already drawn fresh. At α=0.1 it escalated to 765 calls (25 flips in 720) for 90.5% savings. The run applied the loose certificate, persisting a `reuse_certificate` row and stamping its identifier on 79,130 of the 81,289 reused cells, which are what the table then served. The 2,159 it did not stamp already held a freshly computed cell when the certificate was applied, and the apply step refuses to write over one. Under the Bardenet–Maillard bound the loose certificate's evidence (5 flips in 180 draws from the 81,469-cell cached-FALSE stratum) reads 0.230, which does not clear the budget at the look at which it was issued. The exact bound reads 0.073 on the same evidence, which clears α=0.2 and α=0.1 alike. The tight certificate's 25 flips in 720 read 0.055 under the exact bound, which clears α=0.1, and 0.099 under Bardenet–Maillard, which clears it.

c: The August draws under the exact bound.

Replayed over the stored August draws at the loose budget, the exact bound certifies the cached-FALSE stratum at look 45 (flips 1, bound 0.143). The maintenance workload is then 7,760 cells against 89,184 for a full recompute. At α=0.1 it certifies at look 180 (5 flips, bound 0.073) for 225 calls, the count the Maurer–Pontil run needed for the loose budget, reusing 81,289 cells (91.1% savings). At α=0.05 the schedule's deepest look reaches 1,440 rows of the stratum, 699 of which hold no August draw. Realized error was audited against oracle-computed cells of that snapshot only (status `done`/`cached`): the reused rows flip at 3.15%, 89 of 2,821 audited rows. Certificate copies are excluded by construction, as verbatim v1 values would register as non-flips. Of the 2,821 audited reused rows, 1,812 belong to the released n=2,000 evaluation vector and show 3.20% on their own. The released subset is a consistency check with Table 2, and the remainder is independent evidence. Every audited row is released with its cached and oracle values, and the verifier recomputes each audit from that file alone, without a database. Priced at the ledger's measured $0.000118 per cell, the maintenance workload would cost $0.91 and about 42 minutes of wall clock on the latency estimate, against the $9.31 the corpus cost to materialize. That estimate uses the materialization's mean per-call latency at the same 32-way concurrency. We price the recompute leg without executing it.

TABLE S3. The Maurer–Pontil certifications on the deployment column, kept for the record. Theorem 1 does not cover them (Section III). The first row is the certificate the system applied in August 2026. Columns as in Table 6.

Budget	oracle calls	FALSE look / flips	reused	savings	audited error (flips/rows)
August 2026 snapshot: the live run					
α=0.2	225	180 / 5	81,289	91.1%	3.16% (85/2,686) (applied)
α=0.1	765	720 / 25	80,749	90.5%	3.03% (65/2,146)
September 2026 snapshot: the draws of Table 6					
α=0.2	225	180 / 11	81,289	91.1%	7.54% (95/1,260)
α=0.1	1,485	1,440 / 106 (refused)	0	0.0%	–

d: The drift measurement.

The September oracle draws are stored under a version of their own, so draws of the two snapshots never mix. Reasoning effort is left at the provider's default, one of the settings whose change we cannot tell from a change of the model. Of the 720 cached-FALSE rows the August run had sampled, 36 flipped only in September against 11 only in August, the counts behind the McNemar test of Section V-E. A like-for-like comparison separates the model from the edit. The v1 prompt re-drawn in September disagreed with its own August draws, the cache, on 4.4% of those rows (Wilson [3.2, 6.2]). The same prompt's within-snapshot self-disagreement is 2.6% ([2.0, 3.1]). The v2 prompt re-drawn disagreed with its August draw on 6.5% ([4.9, 8.6]). The edit's flip rate with both sides drawn in September was 5.8% ([4.3, 7.8]). The ledger shows the same change: 534 output tokens per cell against 290 in August, at 24 seconds per call against 9. On the 45 cached-TRUE rows the edit's flip rate read 28.9% in August and 24.4% in September. Of those rows, 8 flipped only in August and 6 only in September (McNemar p=0.79). The v2 prompt re-drawn disagreed with its August draw on 31.1% of them ([19.5, 45.7]). The same stratum's floor on the lab column is 23.0%. These rows are too few to resolve a shift.

e: The September certification.

At α=0.2 the certifier certified the cached-FALSE stratum at look 45 (2 flips, against 1 in August) and reused 81,424 cells (91.3%). The cached-TRUE stratum was refused at look 45 with 11 flips. At α=0.1 it escalated to look 360 (20 flips, bound 0.091) and reused 81,109 cells (90.9%). The 315 calls beyond the loose arm's 90 took a further 304 seconds. At α=0.05 the futility rule refused the stratum at look 90 (6 flips, 6.7%), after 135 calls in all, correctly: the stratum's flip rate against the aged cache exceeds that budget. The run issued its oracle calls 32 at a time, the deployment script's setting before the released certifier issued them one at a time (Appendix L). Under Maurer–Pontil the same September draws certify the loose budget after 225 calls and are refused at the tight budget after 1,485 calls (Table S3). The audits cover 1,395 rows under the loose certificate and 1,080 rows under the tight one. The sample path needed 1,485 September oracle cells across the arms. Of these, 699 were already held from a pass earlier that day and 786 were requested from the runner. Of the requested cells, 20 were served by the content-hash cache and 766 were drawn fresh for $0.15. That is $0.000199 per cell against $0.000118 in August. The longer outputs are the difference. The drift measurement drew 752 more.

f: Call counts and reproducibility.

The call count of a certification is itself random, since a schedule stops at the first look whose bound clears α. At α=0.2 the benchmark's main seed spent 90 calls, with a replication median of 90, and at α=0.1 the replications span 90–405 calls (mean 188, median 225). Three reproducibility caveats apply. First, in MySQL the projection list changes which rows a seeded `ORDER BY RAND` returns, so the evaluation sample is defined by an exact query string, not by the seed alone. Second, a linear-congruential generator written with JavaScript number multiplication silently loses low bits, so all sampling uses exact 32-bit arithmetic. Third, the certifier at the time of the August run shuffled each stratum in the order the cell query returned. The query planner served it through one index in August and another afterward. The August run walked the content-hash order, the certifier now sorts by content hash before shuffling, and the replay reproduces every count of the August run under the Maurer–Pontil arm (the released script asserts it).

APPENDIX D THE INDEPENDENCE CHECK IN DETAIL

Table S4 collects the statistics of the check that Section V-F summarizes.

a: The check.

Each of the 300 rows had the formatting edit's new version drawn once with a single request in flight and once at a concurrency of 32 (Section V-F). The sequential arm took 78.3 minutes of wall clock and the concurrent arm 4.6. The planned comparison excludes requests that errored or exceeded a 90-second ceiling, and the ceiling binds more often under concurrency. The released certifier does not exclude them: it counts a failed call as a flip. Under that rule the sequential arm reads 35 flips of 300 and the concurrent arm 52 (two-proportion p=0.049). Of the rows, 21 count as flips only sequentially and 38 only concurrently (exact McNemar p=0.036). A certifier under concurrent load therefore sees more flips than one that issues its calls one at a time, which makes it refuse more often and never certify more. The larger arm, drawn under faster serving, had no failed request. On

TABLE S4. The independence check: one sequential arm and two concurrent arms of the formatting edit's new version, each against the same cache. Lag-one autocorrelation and block dispersion are computed in completion order, the latter over blocks of 32 completions against its binomial value. The launch-order dispersion and the failure-as-flip rate were not in the plan. Flip-rate intervals are 95% confidence intervals.

	Sequential	Concurrent	Concurrent, larger arm
Run date	29 September 2026	29 September 2026	30 September 2026
Rows requested	300	300	1,000
Requests that failed (error or time ceiling)	9	28	0
Flip rate, completed requests (Wilson interval)	8.9% [6.2, 12.8]	8.8% [6.0, 12.8]	7.1% [5.7, 8.9]
Flip rate, failed requests counted as flips	11.7%	17.3%	7.1%
Lag-one autocorrelation (permutation p)	0.070 (0.27)	−0.006 (0.95)	−0.031 (0.34)
Block dispersion ratio (blocks; upper-tail p)	1.39 (9; 0.20)	2.08 (8; 0.04)	1.26 (31; 0.15)
Launch-order dispersion ratio (p)	–	1.82 (0.08)	1.61 (0.019)

completed requests the two arms' rates give a two-proportion p=0.96. On the 263 rows both arms kept, the rates are 9.5% sequential and 8.4% concurrent. Of those rows, 14 flipped only sequentially and 11 only concurrently (exact McNemar p=0.69). The excluded rows show no sign of being flip-prone, on small counts. Of the 28 rows the concurrent arm dropped, 1 flipped in the sequential arm. Of the 9 the sequential arm dropped, 2 flipped in the concurrent arm.

b: The planned tests.

The script fixed three analyses before the run, each reported against its reference distribution at the 5% level. The first is a two-proportion test on the flip rates, whose minimum detectable difference at 80% power is 7.0 percentage points. The second sets the two arms' disagreement on the same row against the August stratum floors. The third looks for serial dependence in completion order, with a lag-one permutation test and a dispersion test of flip counts over consecutive blocks of 32 completions. The two arms disagree with each other on 9.5% of rows ([6.5, 13.7]): 9.2% of 238 cached-FALSE rows and 12.0% of 25 cached-TRUE rows, against August floors of 2.6% and 23.0%. After the drift measured in Section V-E, that second comparison cannot separate concurrency from the change of snapshot, and we draw nothing from it. Table S4 gives the third analysis. In the concurrent arm the dispersion ratio's 95% confidence interval (CI) is [0.91, 8.62], which covers 1, while its upper-tail chi-square test rejects.

c: Reading the tests.

The rate and lag tests do not reject. The one-sided dispersion test rejects at the 5% level in the concurrent arm, on 8 blocks. To add blocks we drew a concurrent arm alone on a further 1,000 rows of the vector. Its block dispersion ratio has a 95% CI of [0.81, 2.26], and its flip rate set against the sequential arm's gives p=0.30. This arm was drawn a day later, on other rows, and under faster serving (a median response of 3.7 seconds against 14.4 in the first concurrent arm), so the comparisons between arms also span a day of the snapshot. Pooled over both concurrent arms, the dispersion statistic is 52.5 on 37 degrees of freedom (p=0.048), a ratio of 1.42 with a 95% CI of [0.94, 2.37]. The block-scale question is therefore open. The larger arm alone does not reject. The pooled test, which includes the first arm, rejects narrowly. The sequential arm's ratio does not reject either, on 9 blocks, which are too few to show that one-at-a-time draws are free of dependence. A block holds about three expected flips, where the chi-square reference is approximate. Permutation p-values for the same statistics, from shuffles of each arm's own flip sequence, differ from the chi-square ones by at most 0.02 and fall on the same side of the 5% level in every case.

Three looks that were not in the plan argue for caution. Flips concentrate in slow responses: in the larger arm, 59 of the 500 slower responses flip against 12 of the 500 faster ones. Completion order is therefore not neutral with respect to the outcome. With blocks formed in launch order instead, the dispersion ratio is 1.61 in the larger arm (p=0.019) and 1.82 in the first (p=0.08). The ordering was chosen after seeing the data, so we report these as cautions, not as findings. The third look asks where the excess of the first concurrent arm comes from. Its flips concentrate late in its completion order: 4 in the first 136 completions and 20 in the last 136 (two-proportion $p < 0.001$). The sequential arm walks the same rows in the same order and leans the same way: 9 flips in its first 145 completions and 17 in its last 146. Of the 263 rows both arms kept, 11 flip in both, where independent rows at those rates would give about 2. The larger arm, on other rows, shows no such trend (34 and 37 flips in its two halves of 500). Block excess in the paired arms is therefore at least partly a property of their rows and the order those rows were issued in. A test in completion order cannot separate that from interference between requests.

d: The simulated cost of dependence.

The simulation behind the error probabilities of Section V-F runs the certifier on the deployment stratum, at its two certified budgets and at a flip rate one flip past the budget, 100,000 times per setting. Draws arrive in blocks of 32 with beta-binomial dependence inside a block, so that block counts have a chosen variance ratio against the binomial. It is a model of dependence, not a measurement of it. Table S5 gives the error probability at every ratio the check reports. With independent draws it is at most 2.3%. At the pooled estimate it is about the nominal level, and at the upper end of the pooled interval it is between two and three times nominal. The first concurrent

TABLE S5. Simulated probability that the certifier errs on a stratum one flip past its budget, by the dispersion ratio of flip counts within blocks of 32 draws (deployment stratum; nominal per-stratum level 5%).

Source of the dispersion ratio	ratio	α=0.2	α=0.1
Independent draws	1.00	2.3%	1.9%
Larger concurrent arm, estimate	1.26	4.1%	3.5%
Both concurrent arms pooled, estimate	1.42	5.4%	4.7%
Larger arm, blocks in launch order (unplanned)	1.61	7.1%	6.1%
First concurrent arm, estimate	2.08	10.8%	9.6%
Larger arm, upper end of its interval	2.26	12.2%	10.8%
Both arms pooled, upper end of the interval	2.37	13.1%	11.7%
First arm, upper end of its interval	8.62	37.9%	38.8%

arm's own interval, from 8 blocks, reaches ratios at which the certifier would err far more often. At the loose budget the deployment stratum is not one whose flip rate sits just above the budget: its audited error is 3.15% in August and 7.46% in September (Table 6). At the tight budget the September audit, 7.96% against α=0.1, leaves less room. The check itself covers one model through one gateway at one concurrency, in the September snapshot. Appendix L gives the change to the released certifier, and the deployment certificates under a bound widened for these ratios.

APPENDIX E THE FREE-TEXT PILOT

a: The recipe.

In a stratum, let q be the share of pairs the judge calls equivalent although a reader would not (missed flips). The flip rate the judge reports for the stratum is then at least the true rate minus q, and q is at most the judge's miss rate among the pairs it calls equivalent. Let $\hat{m}$ be an upper confidence bound on that conditional miss rate at level δ_{cal}, from human-labeled pairs of the same column, and certify at $\alpha - \hat{m}$. A certified stratum's true rate is at most α whenever the calibration bound and the certificate both hold, provided the judge's conditional miss rate in the certified stratum is no higher than in the calibration pool. They both hold with probability at least $1 - \delta - \delta_{\text{cal}}$. Errors in the other direction (calling equivalent cells different) only raise the reported rate and cost power. The calibration belongs to the (edit, column) pair it was drawn from, or to a pool of comparisons on one column if their conditional miss rates are taken as equal, since a judge's miss rate is a property of the outputs it compares.

b: Setup.

The pilot column of Section VIII has three prompt versions: a baseline, a formatting-only edit of it, and a synonym rewording. The judge (`deepseek/deepseek-v4-flash-0731`) is asked in both orders whether two phrases name the same specialty, and a pair counts as equivalent only when both orders say so. A pair is excluded when either of its requests failed (errored or timed out). The failures came in bursts of consecutive rows and did not recur on the same rows across versions. The exclusion leaves 1,602, 1,821, and 1,799 of the 2,000 pairs usable for the floor, the formatting edit, and the synonym edit. Under that judge the column's own floor, two draws of the same prompt, is 40.1% (Wilson [37.8, 42.6]). The formatting edit flips 42.2% ([40.0, 44.5]) and the synonym edit 49.9% ([47.6, 52.2]).

c: Pilot calibration.

The author hand-labeled 201 calibration pairs, drawn evenly from the three comparisons and pooled: 150 the judge had called equivalent and 51 it had called different. Each pair was shown in a shuffled order as two phrases with a Same/Different choice, without the judge's verdict or the comparison it came from. The judge missed 0 true differences among the 150, an upper confidence bound of 2.0% at δ_{cal}=5% on its miss rate among the pairs it calls equivalent. The bound is pooled across the three comparisons; each comparison's 50 labels alone give 5.8%. The judge also called 41 equivalent pairs different. A second model (`meta/muse-spark-1.3-contributor`) labeled the same pairs and agreed with the hand labels on 74.1% of them; the judge itself agreed on 79.6%. The second model called 20 of the 150 judge-equivalent pairs different, every one of them a pair the author called the same. Under its labels the miss bound would be 18.8%, so the clean side of this calibration rests on the author's reading.

d: A second human labeler.

A second labeler, not an author, then labeled the same pairs on a page that showed them in a fresh order, with the same question and without the author's labels or any model's verdicts. The two human labelers agree on 87.1% of the pairs (Cohen's κ 0.39). Of the 52 pairs on which the author and the second model disagree, the second human labeler sides with the model on 20 and with the author on the rest. Among the 150 judge-equivalent pairs, the second human labeler calls 10 different, a miss bound of 11.0%. Counting a pair as different when either human does gives the same bound, since the author called none of those pairs different.

e: Pilot outcome and reading.

Certifying at $\alpha - \hat{m}$ takes 2.0 percentage points off each budget under the author's labels: α=0.2 becomes 0.180 and α=0.1 becomes 0.080. Under the model labeler's labels the deflation would be 18.8 points, more than the tight budget itself, and under the second human labeler's it is 11.0 points, which leaves 0.089 of the loose budget and none of the tight one. Stratification is by the cached phrase's category (8 categories, assigned once from the cache by the same model). In that configuration, both edits are refused on every stratum at both budgets, each run stopping after 360 oracle calls (every stratum at its first look). A larger deflation cannot certify more on the same sample path, so under the second human labeler's bound every stratum stays refused. The composed

guarantee holds with probability $1 - \delta - \delta_{\text{cal}}$ under the assumption stated in the recipe above; the judge's price is the deflation plus the labels. The floor the certifier met is the judge's, not the model's. Under the author's labels, 41 of the 51 judge-different pairs name the same specialty, across the three comparisons, and 0 of the 150 pairs it called equivalent differ. Weighting the judge's verdicts by those label shares puts the disagreement of two same-prompt draws, as the author reads them, near 7.9% (band [4.4, 14.5]% from the Wilson intervals of the two shares), inside the loose budget. As the second human labeler reads them, it is near 24.5% (band [17.3, 32.8]%), above the loose budget at the estimate. The three labelings read the judge differently. By the author's labels it is stricter than the author and misses nothing. By the model labeler's labels it is mostly right when it calls a pair different (42 of 51) and misses 20 of the 150 pairs it calls equivalent. The second human labeler sits between them on both sides, calling 25 of the 51 judge-different pairs the same and 10 of the 150 judge-equivalent pairs different. With the two humans at κ 0.39, the instruction is loose as well as the judge: readers do not agree on which specialties are the same. Either way the certifier, which sees only the judge, refused every stratum. The composed guarantee holds for any judge whose miss rate the calibration bounds correctly. What the pilot does not deliver is a judge whose equivalence matches a reader's, calibrated in both directions.

APPENDIX F THE CALIBRATION STUDY IN DETAIL

a: What a clean stratum pays.

Because the doubling schedule takes the first look at or above n^*, a clean stratum of n_j=5,000 pays 180 draws in calibration. A clean stratum of n_j=600 pays 90, because its finite-population correction and its shorter schedule's larger per-look level bring the earlier look under α.

b: The planted study.

The study of Section V-F plants flip rates $p \in [0, 0.25]$ at stratum sizes $\{600, 1{,}800, 5{,}000\}$ and budgets $\alpha \in \{0.05, 0.1, 0.2\}$, in both modes, with 2,000 trials per configuration. The worst rate of unsafe certificates, 1.00% (95% CI [0.65, 1.54]%), occurs at α=0.05, p=0.0525, n_j=1,800, in the default mode. The presented-cells rate, which leaves out the sampled flips, exceeds α in at most 0.70% of runs. Of the configurations, 54 are planted just above the budget, at 1.05 to 1.25 times α, where the futility rule often cannot dispose of the stratum at the first look and the sampling fraction climbs. Over these the unsafe rate averages 0.35%. Their certification rate reaches 6.2%, and most of those certificates are correct. At p just above α the realized count of a planted stratum falls at or below the budget in a large share of trials, and the exact bound certifies what is actually there. The reuse-set rate evaluated on default-mode certifications fluctuates past α in frontier configurations. The worst is 100.0% of certified runs at p=0.0625, α=0.05, certified 0.80% of the time. That is expected, since the default mode does not certify that metric.

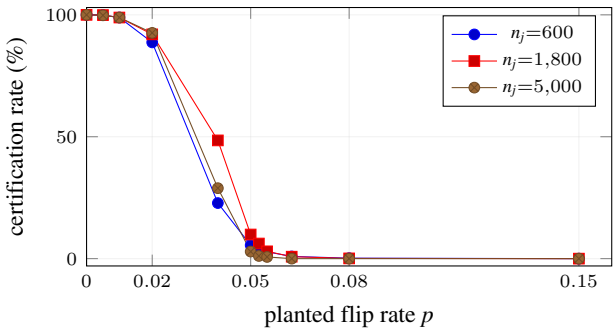


FIGURE S1. Calibration power at α=0.05 (presented-cells estimand, 2,000 trials per point). Certification rises smoothly as p falls below α. Certification is small but nonzero just above it. There the realized population rate of a planted stratum often falls below α, and certifying it is correct. The unconditional rate of certificates whose realized error exceeds α stays at or below 1.00% in every configuration, against the nominal 5%. The n_j=1,800 curve sits above the other two at intermediate p, because the exact bound's finite-population correction tightens as the sampled fraction grows. The six-look schedule reaches most of an n_j=1,800 stratum but a minority of an n_j=5,000 one. The n_j=600 stratum's final look is the whole stratum and cannot certify, so its deepest certifying look is its fourth.

The strict mode holds its own guarantee within δ_j, with unsafe certificates in at most 0.60% of runs.

c: The least favorable population.

Fix a stratum of N cells of which M flip. Under the pinned schedule the certifier's decision at each look is a threshold on the flips seen so far. It certifies if the count is at most the largest count whose bound clears the budget (a look that takes the whole stratum cannot certify). It stops for futility if the sample's rate exceeds the budget, and otherwise it goes on. The flips among each look's new cells are hypergeometric given the flips already seen. The probability of certifying is therefore a finite sum over the paths of cumulative counts, which the released script evaluates exactly, with the thresholds of the released certifier.

That probability does not increase with M. Take two populations that differ in one cell, a flip in the larger and none in the smaller, and sample both by one shuffle. At every look the larger population's sample holds at least as many flips. A path of counts that certifies still certifies when every count is lowered: lower counts pass every threshold the path passed and trip no futility stop that it did not. Every shuffle that certifies the larger population therefore certifies the smaller one. In the default mode a certificate is unsafe exactly when $M > \alpha N$, so its probability is largest at $M = \lfloor \alpha N \rfloor + 1$, one flip past the budget. In the strict mode the unsafe event depends on the look, and the script finds the worst M by scanning. Table S6 gives the result for each combination of the grid. The script also cross-checks one combination per budget by running the released certifier on shuffled populations.

TABLE S6. The exact probability of an unsafe certificate at the least favorable fixed population, for each budget and stratum size of the calibration grid (per-stratum δ_j=5%; values rounded up).

	default mode, n_j			strict mode, n_j		
Budget	600	1,800	5,000	600	1,800	5,000
α=0.05	1.34%	2.27%	1.58%	0.37%	0.87%	0.83%
α=0.1	1.90%	2.60%	2.05%	0.88%	0.75%	1.04%
α=0.2	2.23%	2.31%	2.62%	0.59%	0.76%	1.02%

APPENDIX G THE BUDGETS AND THE STRICT MODE IN DETAIL

The audits of Section V-E put the hidden positives of Section VIII well under what the loose budget tolerates. Carried to all the reused cells, the error rates audited under the loose certificates come to about 2,600 hidden positives on the August draws and about 6,100 in September. The audits are samples, and their 95% confidence intervals carry to 2,100–3,100 and 5,000–7,300.

Table S8 sets the strict mode beside the default mode on the configurations of Table 2, at the three budgets where either certifies. The pattern of Section V-B holds throughout. Where a stratum's flip rate is well below the budget, the two modes agree, as on the three SO edits at α=0.2. One budget step before the default mode's own frontier, the strict mode's savings fall away: on the synonym and scope-widening edits at α=0.1, and on both Djinni edits at α=0.2. The strict mode issues unsafe certificates too, at a rate the theorem allows. Counted per stratum, they occur in 4 configurations of the table, most often on SO scope widening at α=0.05, in 0.4% of replications. Each rate sits inside its column's per-stratum level, δ_j=0.05 on the Boolean columns and δ_j=0.025 on the select column.

TABLE S7. What each budget tolerates and what it buys. Tolerated cells are α times the 81,469 cells of the corpus-scale cached-FALSE stratum, also given as a multiple of the 7,715 rows the column marks TRUE. The corpus-scale columns are the live September certification of Table 6. The last six columns give the certification rate and the mean savings of the formatting-only edit, in percent over 1,000 replications: on the Boolean column in both modes, and on the select column in the default mode (Table 2; Appendix G).

	tolerated, corpus scale		corpus-scale run (September)			Boolean, default		Boolean, strict		select, default	
Budget	cells	× TRUE rows	calls	reused	audited	cert.	savings	cert.	savings	cert.	savings
α=0.2	16,293	2.1	90	81,424	7.46%	100.0	88.1	100.0	88.1	100.0	52.1
α=0.1	8,146	1.1	405	81,109	7.96%	98.9	82.6	98.7	80.6	73.8	8.4
α=0.05	4,073	0.5	135	refused	–	74.4	38.6	4.5	3.4	0.0	0.0

TABLE S8. The default and strict modes on the five edit pairs: certification rate and mean savings over 1,000 replications, in percent. Column *unsafe* is the share of all replications in which the strict mode issues an unsafe certificate, counted per stratum: one for a stratum whose realized flip count exceeds α times the cells it reuses, the event Theorem 1 bounds in that mode.

		default mode		strict mode		
Edit	α	cert.	savings	cert.	savings	unsafe
SO formatting-only	0.05	74.4	38.6	4.5	3.4	0.1
SO formatting-only	0.10	98.9	82.6	98.7	80.6	0.0
SO formatting-only	0.20	100.0	88.1	100.0	88.1	0.0
SO synonym rewording	0.05	0.0	0.0	0.0	0.0	0.0
SO synonym rewording	0.10	81.4	55.6	14.2	11.5	0.0
SO synonym rewording	0.20	99.9	87.4	99.9	87.0	0.0
SO scope widening	0.05	1.2	1.0	0.4	0.3	0.4
SO scope widening	0.10	88.8	66.3	37.6	29.5	0.0
SO scope widening	0.20	99.9	86.6	99.9	86.4	0.0
Djinni formatting-only	0.05	0.0	0.0	0.0	0.0	0.0
Djinni formatting-only	0.10	73.8	8.4	0.8	0.2	0.0
Djinni formatting-only	0.20	100.0	52.1	91.1	29.7	0.2
Djinni criteria change	0.05	0.0	0.0	0.0	0.0	0.0
Djinni criteria change	0.10	82.8	14.4	2.7	0.8	0.0
Djinni criteria change	0.20	100.0	48.6	97.6	31.6	0.2

APPENDIX H THE BASELINES IN DETAIL

a: B1 and B2.

On the formatting edit the similarity score's entire range is 0.96–0.99, because shared row text dominates. The aggregate-only arm runs the same schedule and budgets as the pinned procedure. On the formatting edit it certifies at α=0.1 (82.0% savings, 6.46% aggregate error) while the reused cached-TRUE subgroup carries 35.3% error. On the widening edit it refuses at α=0.1 and certifies 95.5% savings at α=0.2 with 44.6% subgroup error. The effect is not an artifact of giving the baseline a shorter schedule.

b: B2′: how the competitor was matched.

We matched the competitor to sIVM on every axis we could: the same number of oracle calls, identical labels, identical error target, the same exact bound on the selected population, the same total failure probability δ, and the same reported quantity. Each method's δ is split across the number of tests its own procedure performs: sIVM across the Boolean column's two strata and six looks, $\delta/12$ per test; the competitor across its candidate thresholds, $\delta/3$. The reported quantity is the error among reused cells, not the presented-cells rate sIVM certifies. Run instead on each stratum separately at δ/K with a single look, the competitor would be the non-adaptive special case of sIVM itself.

At α=0.2 on the formatting edit the competitor reused 1,910 cells to sIVM's 1,769 from the same 90 labels. At 6.54% realized error it satisfies the bound it was given, against sIVM's 3.28%. On the widening edit the same pattern holds: 1,910 reuses at 9.58% versus 1,756 at 5.47%. The extra volume is drawn from the volatile stratum, 183 cached-TRUE cells on the formatting edit and 196 on the widening edit. With the cached value as the proxy there are only three candidate thresholds, and at α=0.2 the aggregate bound clears at the loosest one. The arm therefore selects 100.0% of its allowed rows on the formatting edit and 100.0% on the widening edit. Matched this way, the competitor reduces to B3, reuse everything, with a certificate attached. Its bound is satisfied, its aggregate error is real, and its behavior is indistinguishable from the guarantee-free status quo.

c: B2′: two caveats.

First, the cached value is the proxy the reduction names, and the result does not depend on that choice. Run instead with an embedding-interaction proxy, and with its much larger threshold search corrected accordingly, the competitor certifies in 3 of the 4 (edit, budget) configurations and lands in the same place. At α=0.2 on the formatting edit it reuses 1,899 cells at 6.53% error, 180 of them cached-TRUE cells that flip at 37.2%. On the widening edit it reuses 1,856 at 9.21%, 168 of them cached-TRUE cells at 50.6%. The proxy changes how much the competitor reuses, not where its errors land. Second, with the cached-value proxy the competitor's per-test level is looser than sIVM's, because it runs fewer tests. The embedding arm, tested at a stricter level than sIVM's, still reuses more than sIVM at α=0.2, so that advantage does not rest on the looser level. The comparison is matched on the quantity a user sets (δ in total), not on the level of any individual test.

d: B3: the threshold sweep.

The cells the embedding-transfer baseline reuses at τ=0.95 flip at 6.35% on the formatting edit and at 9.8% on the widening edit, where the population rate is 9.45%. A sweep of τ over the per-row similarities shows the baseline collapsing within a threshold window about 0.03 wide. On the formatting edit it falls from 100% of rows reused at τ=0.96

to 10.7% at 0.98 and none at 0.99. On the widening edit it falls from 99.9% at 0.94 to 8.1% at 0.96 and none at 0.97. The realized error among reused rows stays at the population flip rate at every operating point (5.6%–6.6% and 8.0%–9.8%). The threshold controls volume, never error, because cross-version similarity carries no per-cell flip signal (B1).

APPENDIX I THE GRADED RELATION IN DETAIL

a: The re-scoring.

Exact match is the strictest relation a select column can use, and the relation is the user's to state (Section VIII). Every self-flip observed on the seniority column was between adjacent tiers. After the exact-match results were in, we re-scored the stored labels of that column's formatting edit under a tolerance: a cell flips only when the fresh tier is more than one step from the cached one. Under the tolerance the edit flips 4 of 2,000 cells, where exact match flips 12.30%. The pinned procedure then certifies it in every replication at α=0.05, a budget that exact match refuses, for 66.3% mean savings. This is a weaker guarantee, not a tighter one. It bounds moves of two tiers or more and says nothing about moves of one tier. The certifier is unchanged: Theorem 1 needs only a fixed binary flip indicator, which need not come from a transitive relation. The tolerance would be the wrong relation for the column's criteria edit, whose stated purpose is to move some candidates up one tier. Of that edit's 249 exact-match flips, 244 are one-tier moves, and a tolerance would certify away the change the user asked for.

APPENDIX J FURTHER TABLES AND ANALYSES

Five tables the article cites are printed here, with three further analyses: two of the model study and one of the minority stratum.

a: Models and the headline pair's flip rates.

Table S9 lists every model the article uses, by gateway identifier, with the gateway's list prices. Table S10 reconciles the flip-rate figures quoted for the headline pair and its cached-TRUE stratum (Section V-A).

b: The remaining pairs on further models.

Table S11 extends the model study of Section V-D to the three edit pairs that Table 5 does not cover.

c: The formatting edit on two further models.

On Gemini 2.5 Flash-Lite the formatting edit flips 25.45% of cells against a floor of 4.85%. Decomposed, its two paragraph breaks alone flip 23.2% and its one-word case change alone 11.0%. The edit also moves the share of TRUE answers on that model, from 21.8% to 34.8%. On GLM-4.7-Flash it moves that share the other way, from 80.0% to 61.8%. Self-disagreement alone flips cells both ways and leaves the share where it was, so on these two models the edit shifts the model's decision.

d: Floors are per stratum.

The floor binds for the identity edit only (Section II), and some strata of the model study (Section V-D) flip less under an edit than under none. The two clearest are on the lab column. Under the formatting edit GLM-4.7-Flash's cached-FALSE stratum flips 17.0% against an identity floor of 24.4%, and GPT-5 nano's cached-TRUE stratum flips 24.3% against 44.9%. GPT-5 nano's pooled floor, 34.77%, sits above every budget in the grid. Its one certifying stratum on the lab column sits below that pooled figure for the same reason: a floor is a per-stratum quantity, measured under the first version's prompt.

e: The minority stratum under majority voting.

The cached-TRUE stratum of the Boolean column flips 37.1% under a formatting edit. One candidate cause is cache noise, which a majority vote of three draws at write time would remove [24]. The stored draws do not support it. Voting on both sides (vote-of-3) brought the stratum's flip rate to 24.7% in a dedicated probe and to 34.4% on the three-draw labels, two separate runs over the same 186 rows. The vote overturned 20.4% and 16.7% of the cached TRUE values, respectively. A cache built by the vote fares no better. Its TRUE stratum, the 188 rows whose three-draw majority is TRUE, has a lower floor (17.6% against 23.0%) but flips at 41.0% against fresh draws of the edited prompt (Table S10). Voting at write time cleans the cache without making the minority stratum certifiable under this edit. Refusing and recomputing that stratum is the response the data support.

f: The benchmark's edit pairs.

Table S12 transcribes the five versioned prompt-edit pairs of the benchmark (Section VI) from the stored prompt templates.

g: Neighboring systems compared.

Table S13 sets sIVM against the lines of work it borrows from, is compared with, or could be mistaken for, by the quantity each certifies and the axis of change it holds fixed (Section VII).

TABLE S9. Every model the article uses, by gateway identifier, with the gateway's list prices as fetched on 29 September 2026.

Gateway identifier	Role	Run dates	$/M tokens in / out
`deepseek/deepseek-v4-flash-0731`	primary model: every cell of the main program, the deployment run, the drift and independence checks, the free-text pilot's cells and its judge	4 August 2026 to 5 August 2026; 29 September 2026; 30 September 2026	0.08 / 0.15
`openai/text-embedding-3-small`	embedding proxy (baselines B1, B3; interaction stratifier)	with the main program	0.02 / 0.00
`google/gemini-2.5-flash-lite`	model study, all five pairs	15 August 2026; 29 September 2026	0.10 / 0.40
`google/gemini-3-flash`	model study, lab column's two pairs	15 August 2026	0.50 / 3.00
`zai/glm-4.7-flash`	model study, all five pairs	15 August 2026; 29 September 2026	0.07 / 0.40
`alibaba/qwen3.7-flash`	model study, all five pairs	15 August 2026; 29 September 2026	0.03 / 0.13
`openai/gpt-5-nano`	model study, all five pairs	29 September 2026	0.05 / 0.40
`google/gemini-3.8-flash`	model study, all five pairs	29 September 2026	0.75 / 3.75
`meta/muse-spark-1.3-contributor`	free-text pilot: second labeler of the calibration pairs, reported beside the hand labels	30 September 2026	0.10 / 0.20

TABLE S10. Reconciliation of the flip-rate figures quoted for the headline pair on the evaluation vector and for its cached-TRUE stratum. Each is a different quantity, a different instrument on the same quantity, or the same instrument on an earlier draw; none is a restatement of another.

Rate	Edit	Cells	Instrument (artifact)
Headline pair: SO formatting-only, Boolean lab column			
6.35%	formatting v1→v2	n=2,000 evaluation vector	stored v2 oracle cells against the stored v1 cache, one draw each side (127 flips); the Table 2 label instrument (`exp8`)
6.0%	formatting v1→v2	same rows	the same comparison on the v2 cells as first drawn (120 flips), recorded before 1,811 of them were re-drawn on 5 August 2026 (`exp10`)
6.30%	formatting v1→v2	same rows	majority of three draws on each side (126 flips; `exp10`)
7.2%	formatting v1→v2	same rows	fresh single draws on both sides, neither the cache (`exp10`)
5.0%	identity (no edit)	same rows	pairs of fresh draws of one version: the column-level self-flip rate (`exp10`)
2.6%, 23.0%	identity	cached-FALSE, cached-TRUE rows	the stratum floors: the stored v1 value against each of three fresh v1 draws (4.5% pooled; `exp10`)
Cached-TRUE stratum figures			
37.1%	formatting v1→v2	cached-TRUE rows of the vector	single fresh draw against the cache (`exp4`)
24.7%	formatting v1→v2	same	vote-of-3 on both sides (`exp4`)
34.4%	formatting v1→v2	same	vote-of-3 on both sides, recomputed from the three-draw labels of a separate run (`exp10`)
17.6%	identity	188 rows whose three-draw v1 majority is TRUE	the stored v1 value against that majority: the floor of a cache built by vote (`exp10`)
41.0%	formatting v1→v2	same	each fresh v2 draw against that majority (`exp10`)
45.7%	widening v1→v2, production column	199 cached-TRUE rows	single fresh draw against the cache (Wilson [39.0, 52.7])
35.3%	formatting v1→v2	cached-TRUE cells B2 reused at α=0.1	realized error among the cells the aggregate-only baseline reused, an adversely selected subset, not a population rate (`exp7`)
37.2%	formatting v1→v2	same at α=0.2	same (`exp7`)
44.6%	widening v1→v2, production column	cached-TRUE cells B2 reused at α=0.2	same (`exp7`)

TABLE S11. The remaining three edit pairs on further models (same protocol and same entries as Table 5). Floors are each model's self-flip rate on that column, measured in the same run; the primary model's select-column floor is the n=200 probe of the noise-floor paragraph. The primary row repeats Table 2. Its widening-column floor is not shown, because the primary model's self-flip rate under the production column's first version was not measured (the lab column's 5.0% is under the widened definition). A stratum smaller than the first look is drawn in full and cannot be certified: Gemini 2.5 Flash-Lite holds 5 cached-TRUE cells on the widening column.

	SO scope widening (production column)				Djinni seniority tier (four-way select)						
					formatting-only (v1→v2)				criteria change (v2→v3)		
Model	floor	flip	α=0.1	α=0.2	floor	flip	α=0.1	α=0.2	flip	α=0.1	α=0.2
DeepSeek V4 Flash (primary)	–	9.45%	66.3 (88.8)	86.6 (99.9)	6.50%	12.30%	8.4 (73.8)	52.1 (100.0)	12.45%	14.4 (82.8)	48.6 (100.0)
Qwen3.7-Flash	1.50%	17.15%	0.1 (1.5)	24.3 (99.3)	3.10%	5.85%	54.7 (99.9)	83.5 (100.0)	6.60%	39.9 (98.2)	83.2 (100.0)
Gemini 2.5 Flash-Lite	0.00%	8.60%	21.1 (45.8)	94.2 (99.5)	0.15%	6.47%	30.5 (99.9)	85.3 (100.0)	5.07%	53.5 (100.0)	86.1 (100.0)
Gemini 3.8 Flash	0.90%	4.60%	82.4 (95.9)	94.3 (100.0)	3.10%	4.15%	57.6 (100.0)	86.4 (100.0)	11.45%	24.9 (91.6)	56.4 (100.0)
GPT-5 nano	5.75%	36.55%	0.0 (0.0)	0.0 (0.0)	16.25%	15.80%	0.0 (0.1)	29.3 (99.8)	21.10%	0.0 (0.0)	18.7 (98.2)
GLM-4.7-Flash	9.40%	17.95%	46.6 (81.3)	72.6 (100.0)	4.41%	8.46%	20.1 (82.7)	80.4 (100.0)	9.31%	38.7 (99.9)	66.9 (100.0)

TABLE S12. The five versioned prompt-edit pairs. Deltas are transcribed from the stored prompt templates; referenced fields are the row attributes the template interpolates. Flip rates are single-fresh-draw labels against the cached version on the n=2,000 evaluation vector (Table 2); the SO synonym pair is measured against the formatting version's cells (v2→v3), as the benchmark defines it.

Pair	Column (corpus; contract; referenced fields)	Exact delta	flip
SO formatting-only (v1→v2)	Data-platform specialist, lab replica (SO; Boolean; role, databases, platforms, tools, other tech)	Two paragraph breaks inserted after the question and before the "Judge from" sentence; "OR heavy analytics engineering" lowercased to "or heavy analytics engineering"; no content word changed	6.35%
SO synonym rewording (v2→v3)	same	"this person" → "this individual"; "works primarily on" → "chiefly builds and maintains"; "heavy analytics engineering" → "intensive analytics engineering"; "Judge from role … and other tech" → "Decide using their role … and other technologies"; "Working with multiple … counts as evidence" → "Experience with several … is supporting evidence"	8.90%
SO scope widening (v1→v2)	Data-platform specialist, production column (SO; Boolean; same fields)	Definition extended from "warehouses, lakes, pipelines, distributed data stores, or ML data tooling" by "OR heavy analytics engineering"; the exclusion "Merely using one relational database for app work does not qualify" replaced by the inclusion "Working with multiple analytical/distributed databases (BigQuery, Snowflake, Spark, Kafka, Elasticsearch) counts as evidence"	9.45%
Djinni formatting-only (v1→v2)	Seniority tier, lab (Djinni; four-way ordinal select; position, years of experience, CV text)	The one-line template broken into a question line, three labeled input lines (Position, Years of experience, CV), and the instruction sentence, with no word changed	12.30%
Djinni criteria change (v2→v3)	same	Sentence appended: "Evidence of team or people leadership (leading engineers, owning hiring, running a team) moves the candidate up one tier"	12.45%

TABLE S13. What neighboring systems certify, and across what. sIVM is the only row whose certified quantity is the reuse of cached outputs across a change of the prompt definition.

System or line	Certified quantity	Held fixed	Changes	Guarantee form
SUPG, ABae, BARGAIN, LOTUS, ThalamusDB, Task Cascades [11], [26]–[30]	precision, recall, or accuracy of one query's answer against an oracle	the query definition	nothing (one-shot execution)	finite-sample bound per query
vCache [41]	correctness of serving a cached answer to a *new* query	the answer semantics	the incoming query	learned per-entry threshold at a user error rate
FreshCache, Evergreen [42], [43]	probability that a cached entry is stale (world drift); cost of verifying aggregates	the query definition	the world the answer describes	fitted staleness model per tier; confidence sequences
Stale-view cleaning, online aggregation [44], [45]	error of an answer served from a stale view; a running aggregate	the view definition	the base data	sampling bound on the answer
Classical incremental view maintenance (IVM) [46]	exact equality of the maintained view with recomputation	the view definition	the base data	algebraic (no sampling)
View adaptation after redefinition [47]	exact equivalence of the rewritten view	deterministic operators	the view definition	algebraic (no sampling)
Step-level reuse [51]	per-step equivalence under a perturbed request	the model	the request	deterministic per-step check
Guarantee-free semantic caches [20] (B3)	nothing: a hit is a similarity above a threshold	nothing	the incoming query	none
Per-output verification: assertions [54], regression suites [55], majority vote [24]	correctness of one output against a checker or a majority	the prompt	the output under test	per-output check, no population bound
Conformal and distribution-free risk control for model outputs: risk-controlling prediction sets, Learn then Test [15], [32]	a risk of one fixed predictor's outputs on new inputs	the predictor (model and prompt)	the input	finite-sample bound from a held-out calibration set
sIVM (this work)	false-reuse rate of cached stochastic cells kept under the new definition, per stratum	the model snapshot and the data	the prompt definition	exact finite-sample bound per stratum and look

APPENDIX K LABEL NOISE IN THE REPLICATIONS

The replications of Table 2 rerun the certifier with 1,000 sampling seeds over one frozen set of labels, so their intervals show sampling variability only. A label is itself one fresh draw of the edited prompt, and the three-draw relabeling of the headline pair (Section V-A) holds three further draws per row. Only that pair was relabeled. Table S14 reruns the same replication analysis with each of them as the label, against the same cache. The pair's flip rate ranges from 6.15% to 6.55% across the four labelings. At α=0.1 and α=0.2 the mean savings move by less than 1.5 points, because the cached-FALSE stratum's flip rate stays far below both budgets under every labeling. At α=0.05 that rate sits close to the budget: the certification rate ranges from 66.6% to 82.0% and the mean savings from 28.6% to 51.3%. Near a stratum's own flip rate the frontier depends on the draws behind the labels; well above it, it does not. The select column's certified strata sit nearer their budgets, so its label noise is likely larger, and it is not measured here.

TABLE S14. The replication analysis of the headline pair (Table 2, presented-cells estimand) uses the stored labels of Table 2 and each fresh draw (draws 1 to 3) of the three-draw relabeling as the label. It reports flip rates overall and per cached-value stratum, and mean savings with the certification rate in parentheses, in percent over 1,000 replications. Draw 3 is missing for one row, so that labeling covers 1,999 rows.

	flip rate			mean savings (certification rate)		
Labels	all	FALSE	TRUE	α=0.05	α=0.1	α=0.2
stored	6.35%	3.2%	37.1%	38.6 (74.4)	82.6 (98.9)	88.1 (100.0)
draw 1	6.50%	3.0%	40.3%	45.7 (78.6)	82.9 (98.8)	88.1 (100.0)
draw 2	6.55%	3.5%	36.0%	28.6 (66.6)	81.3 (98.7)	88.0 (100.0)
draw 3	6.15%	2.8%	38.7%	51.3 (82.0)	83.7 (99.1)	88.2 (100.0)

APPENDIX L ONE CALL AT A TIME, AND A WIDENED BOUND

The independence check of Appendix D could not rule out dependence among oracle draws issued together. This appendix gives the two responses of Section V-F: a change to the released certifier, and the reported certificates under a bound widened for the measured dispersion.

a: One call at a time.

The released certifier issues a certification's oracle calls one at a time (Section IV). This costs wall clock and no calls. The independence check's sequential arm took 15.7 seconds per call on the September snapshot. At that rate the loose certificate's 90 calls take about 24 minutes, against 101 seconds at 32 at a time; the tight certificate's 405 take about 1.8 hours, and a run that exhausts the schedule, 1,485 calls, about 6.5 hours. A longer window carries a risk of its own: a change of the served model during a certification shifts every later draw at once, a common shock that the sequential arm, on 9 blocks, was too short to detect. That arm is also too short to show that one-at-a-time draws are free of dependence (its dispersion ratio was 1.39), so the change narrows the assumption's exposure without removing it.

b: A bound widened for dispersion.

For certificates whose draws were issued together, the bound can be widened for a dispersion ratio ϕ. Korn and Graubard's interval for survey proportions [21] puts an effective sample size (the sample size divided by the design effect) and the observed rate times it into the Clopper–Pearson limits. We take the design effect to be ϕ, so k flips in n draws count as k/ϕ flips in n/ϕ. Their interval also shrinks the effective sample size by a degrees-of-freedom factor, for the noise in an estimated design effect. We leave it out of the bound and charge for that noise instead by also evaluating the bound at the upper end of the ratio's 95% confidence interval; the factor's own effect is reported below. The beta form of the limit is defined for non-integer counts. At $\phi = 1$ it is the binomial Clopper–Pearson bound, which agrees with the exact bound to three decimal places at the deployment stratum's size (Appendix B). Table S16 runs the certifier with that bound through the simulation of Table S5. Widened for the ratio that generates the draws, its error probability is at most 2.0% at

either budget, inside the nominal level. The simulation fixes one stratum size, a rate one flip past the budget, two budgets, blocks of 32, and two ratios, so this reading holds in those settings. Widened for the pooled estimate, 1.42, while the draws sit at the upper end of its interval, 2.37, it reaches 7.2%. The widening protects against the dependence it is told about, and not against more.

Table S15 replays the deployment certificates of Table 6 under the widened bound, over the same stored sample paths. At the pooled estimate the loose September certificate needs 135 calls instead of 90 and reuses 81,379 cells; at the upper end it needs 225 and reuses 81,289. The tight September certificate holds at the pooled estimate at its original cost, by a margin of less than a thousandth (its bound is 0.0998), and is refused at the upper end after 1,485 calls. Both August certificates hold at both ratios. The last column of Table S15 gives the largest ratio at which each certificate is still issued on its stored path, and the look at which it then is. The tight September certificate tolerates 1.42 at the look it was issued and 1.97 at look 1,440. With Korn and Graubard's degrees-of-freedom factor, at the 37 degrees of freedom of the pooled estimate and the schedule's deepest look, where the factor is largest, the pooled estimate acts as a ratio of 1.55. Replayed at that ratio, the tight September certificate moves to that deepest look (1,485 calls) and no other certificate changes. The evidence of the certificate the system applied in August, 5 flips in 180 draws, gives widened bounds of 0.085 and 0.111 against α=0.2, and clears that budget up to a ratio of 6.08.

TABLE S15. The deployment certificates of Table 6 replayed under a bound widened for a dispersion ratio of flip counts: the oracle calls of each certificate, or the calls spent before a refusal (ref.). Ratio 1.00 is the exact bound of Table 6; the others are the independence check's pooled estimate and the upper end of its 95% confidence interval. The August draws do not reach α=0.05. The last column is the largest ratio at which the certificate is still issued on its stored path, and the look at which it then is.

		dispersion ratio			largest
Snapshot	Budget	1.00	1.42	2.37	ratio (look)
Aug.	α=0.2	90	90	135	22.30 (720)
Aug.	α=0.1	225	225	405	6.31 (720)
Sep.	α=0.2	90	135	225	25.56 (1,440)
Sep.	α=0.1	405	405	ref. (1,485)	1.97 (1,440)
Sep.	α=0.05	ref. (135)	ref. (135)	ref. (135)	–

c: What a widened certificate asserts.

The widened bound is a binomial limit with no finite-population term. It bounds the probability that a recompute flips a cell of the stratum, under the block model, and not the realized count M_j that Theorem 1 certifies; under interference the coupling that defines M_j fails. At every look where a widened certificate is issued here, the widened bound sits at or above the exact bound, so on the event of Theorem 1 such a certificate clears only a stratum that meets its budget. The theorem says nothing about the protection the widening adds under dependence. Three limits apply to that protection. The replay is retrospective, since the ratios come from a check run after the draws. The ratio was measured on the September snapshot and is applied to the August draws as well, which were served at 9 seconds per call against 24 in September. And the uncertainty in ϕ is not charged to δ: a statement at the upper end of the ratio's 95% confidence interval holds, approximately and in the simulated settings, at level $\delta + 0.05$.

TABLE S16. Simulated probability that the certifier errs on a stratum one flip past its budget (deployment stratum; draws in blocks of 32 at the given dispersion ratio), with the bound as pinned and widened for an assumed ratio (nominal per-stratum level 5%).

	α=0.2, draws at ratio			α=0.1, draws at ratio		
Bound	1.00	1.42	2.37	1.00	1.42	2.37
as pinned (Table S5)	2.3%	5.4%	13.1%	1.9%	4.7%	11.7%
widened for 1.42	0.6%	2.0%	7.2%	0.5%	1.6%	6.0%
widened for 2.37	0.0%	0.2%	1.6%	0.0%	0.1%	1.0%